

A system has RAM which can store four pages simultaneously to satisfy page requests. Write a program to calculate percentage page hit and page fault if the system uses Optimal page replacement technique for the page references entered by the user i.e. 1, 2,3,4, L,2,5, !,2,3,4, 5 (No of Frames=3).

CODE:

```
GNU nano 6.2 optimal.c *
#include <stdio.h>

int main() {
    int frames[10], pages[50];
    int n, f, i, j, k, pos, faults = 0, hits = 0;
    int farthest, index;

    // Input
    n = 12;
    int tempPages[12] = {1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5};
    for(i = 0; i < n; i++)
        pages[i] = tempPages[i];

    f = 3;

    // Initialize frames
    for(i = 0; i < f; i++)
        frames[i] = -1;

    for(i = 0; i < n; i++) {
        int found = 0;

        // Check if page is already in frame
        for(j = 0; j < f; j++) {
            if(frames[j] == pages[i]) {
                hits++;
                found = 1;
                break;
            }
        }

        if(found == 0) {
            faults++;

            // Find optimal page to replace
            farthest = i;
            index = -1;

            for(j = 0; j < f; j++) {
                int foundFuture = 0;
                for(k = i + 1; k < n; k++) {
                    if(frames[j] == pages[k]) {
                        if(k > farthest) {
                            farthest = k;
                            index = j;
                        }
                    }
                }
            }

            // If page not used in future
            if(foundFuture == 0) {
                index = j;
                break;
            }
        }

        // If all pages will be used, replace farthest
        if(index == -1)
            index = 0;

        frames[index] = pages[i];

        // Display frames
        printf("\nStep %d: ", i + 1);
        for(j = 0; j < f; j++) {
            if(frames[j] != -1)
                printf("%d ", frames[j]);
            else
                printf("- ");
        }
    }
}
```

```
GNU nano 6.2 optimal.c *
        faults++;

        // Find optimal page to replace
        farthest = i;
        index = -1;

        for(j = 0; j < f; j++) {
            int foundFuture = 0;
            for(k = i + 1; k < n; k++) {
                if(frames[j] == pages[k]) {
                    if(k > farthest) {
                        farthest = k;
                        index = j;
                    }
                }
            }

            // If page not used in future
            if(foundFuture == 0) {
                index = j;
                break;
            }
        }

        // If all pages will be used, replace farthest
        if(index == -1)
            index = 0;

        frames[index] = pages[i];

        // Display frames
        printf("\nStep %d: ", i + 1);
        for(j = 0; j < f; j++) {
            if(frames[j] != -1)
                printf("%d ", frames[j]);
            else
                printf("- ");
        }
    }
}
```

```
GNU nano 6.2 optimal.c *
    }
    foundFuture = 1;
    break;
}
}

// If page not used in future
if(foundFuture == 0) {
    index = j;
    break;
}

// If all pages will be used, replace farthest
if(index == -1)
    index = 0;

frames[index] = pages[i];

// Display frames
printf("\nStep %d: ", i + 1);
for(j = 0; j < f; j++) {
    if(frames[j] != -1)
        printf("%d ", frames[j]);
    else
        printf("- ");
}

// Output
printf("\n\nTotal Page Faults = %d", faults);
printf("\nTotal Page Hits = %d", hits);

float hitRatio = (float)hits / n * 100;
float faultRatio = (float)faults / n * 100;

printf("\nHit Percentage = %.2f%%", hitRatio);
printf("\nFault Percentage = %.2f%%", faultRatio);

return 0;
}
```

OUTPUT:

```
who@Waghmare08: ~
who@Waghmare08:~$ nano optimal.c
who@Waghmare08:~$ gcc optimal.c -o optimal
who@Waghmare08:~$ ./optimal

Step 1: 1 - -
Step 2: 1 2 -
Step 3: 1 2 3
Step 4: 1 2 4
Step 5: 1 2 4
Step 6: 1 2 4
Step 7: 1 2 5
Step 8: 1 2 5
Step 9: 1 2 5
Step 10: 3 2 5
Step 11: 4 2 5
Step 12: 4 2 5

Total Page Faults = 7
Total Page Hits = 5
Hit Percentage = 41.67%
Fault Percentage = 58.33%who@Waghmare08:~$ |
```